

Human Review: nn-Bounded Operators and nn-Convergence

Original automatic proof: `solution_packet.pdf`

Checked date: 15 June 2026

Status: Rejected as literature already answered.

Verdict: This packet should not be counted as a novel counterexample. The question was answered in later literature, specifically in Theorem 2.2 of Zabetti's paper *Topological Algebras of Bounded Operators on Topological Vector Spaces*.

Human Comments

The automatic packet addresses Troitsky's question asking whether the class of nn-bounded operators is closed under nn-convergence. The proposed construction uses finite truncations of the left shift on $\mathbb{R}^{\mathbb{N}}$, with the product topology, converging in nn-convergence to the full left shift, which is not nn-bounded.

The mathematical construction is valid, but it should be excluded from the survey as a new discovery, because the same phenomenon is already treated in the literature:

Zabetti, *Topological Algebras of Bounded Operators on Topological Vector Spaces*,

Journal of Advanced Research in Pure Mathematics 3(1) (2011), 22–26.

The DOI metadata for this paper is

10.5373/jarpm.438.052210.

More specifically, Theorem 2.2 of that paper states that, for a locally pseudo-convex topological vector space X , the operations in $B_{nn}(X)$ are continuous for the topology of nn-convergence, and that $B_{nn}(X)$ fails to be closed in this topology. The proof gives the same kind of example: take $X = \mathbb{C}^{\mathbb{N}}$ with the topology of coordinatewise convergence, let S_n be the left shift followed by projection onto the first n coordinates, and observe that S_n nn-converges to the full left shift S , while S is not nn-bounded, as in Troitsky's Example 2.5.

Thus this packet is best classified as *already answered by later work*, rather than as a new counterexample produced by the pipeline. It appears to be a rediscovery of the example in Zabetti's Theorem 2.2. I would therefore reject it for inclusion among novel or potentially novel mathematical outputs.

Review Outcome

Rejected. The reason is not that the construction is incorrect, but that the question was already answered in the later literature cited above. This should be moved conceptually to the literature-already-answered category if the repository classification is updated later.